

dark matter Matter that exerts a gravitational influence on its surroundings but does not emit detectable amounts of radiation. Dark matter appears to make up a large fraction of the total amount of mass contained in galaxies, galaxy clusters, and the universe as a whole.

dark nebula see *nebula*.

declination The angular distance of a celestial body north or south of the celestial equator. Declination is positive (+) if the object is north of the celestial equator and negative (–) if it is south of the celestial equator. A star on the celestial equator has a declination of 0°, whereas a star at one of the celestial poles has a declination of 90°. See also *celestial equator*, *celestial sphere*, *right ascension*.

declination axis see *equatorial mounting*.

diffuse nebula A luminous cloud of gas and dust. The term “diffuse” refers to the cloud’s fuzzy appearance and to the fact that it cannot be resolved into individual stars. See also *nebula*.

direct motion see *retrograde motion*.

direct rotation see *retrograde rotation*.

Doppler effect The observed change in the wavelength or frequency of radiation that is caused by the motion of its source toward or away from an observer. See also *blue shift*, *red shift*.

double star Two stars that appear close together in the sky. If the two stars revolve around each other, the system is called a binary. An **optical double star** consists of two stars that appear to be close together only because they happen to lie in almost exactly the same direction when viewed from Earth; they lie at different distances and are not physically connected. See also *binary star*.

dwarf planet A celestial body that orbits the Sun and has sufficient mass and gravity to be spherical, but has not cleared the region around its orbit of other bodies, and is not a satellite.

dwarf star An alternative name for a main-sequence star that was originally devised to distinguish main-sequence stars, such as the Sun, from the much more luminous giant stars on the Hertzsprung–Russell diagram. See also *Hertzsprung–Russell diagram*, *main sequence*.

E

eccentricity (e) A measure of how much an ellipse deviates from a perfect circle. Eccentricity takes a value between 0 and 1; a circle has eccentricity of 0, and the most elongated ellipses approach an eccentricity of 1. See also *ellipse*.

eclipse The passage of one celestial body into the shadow cast by another. A **lunar eclipse** occurs when the Moon passes into Earth’s shadow and a **solar eclipse** when part of Earth’s surface enters the shadow cast by the Moon. A **total lunar eclipse**

takes place when the whole of the Moon lies within the dark cone of Earth’s shadow, and a **partial lunar eclipse** when only part of the Moon is in the shadow. During a **total solar eclipse**, the Sun is completely obscured by the dark disk of the Moon. A **partial solar eclipse** occurs when only part of the Sun’s surface is hidden. If the Moon passes directly between the Sun and Earth when it is close to apogee, it will appear smaller than the Sun, and its dark disk will be surrounded by a ring, or annulus, of sunlight; an event of this kind is called an **annular eclipse**. See also *apogee*.

eclipsing binary see *binary star*.

ecliptic The track along which the Sun appears to travel around the celestial sphere, relative to the background stars, in the course of a year. It is equivalent to the plane of Earth’s orbit.

ejecta Material thrown outward by the blast of an impact. Ejecta, which is produced when a meteorite strikes the surface of a planet or moon and excavates a crater, consists of freshly exposed material that may be markedly brighter than the adjacent surface. Sometimes the ejected material forms extensive streaks, or rays, which radiate from the point of impact. An **ejecta blanket** is a continuous sheet of deposited ejecta that surrounds a crater. See also *crater*.

electromagnetic (EM) radiation

Oscillating electric and magnetic disturbances that propagate energy through space in the form of waves (electromagnetic waves). Examples include light and radio waves.

electromagnetic spectrum The complete range of electromagnetic radiation from the shortest wavelengths (gamma rays) to the longest wavelengths (radio waves).

electron A lightweight fundamental particle with negative electrical charge. A cloud of electrons surrounds the nucleus of an atom. The number of orbiting electrons in an atom is the same as the number of protons in its nucleus.

ellipse An oval curve drawn around two points called **foci** (singular: **focus**) such that the total distance from one focus to any point on the curve and then back to the other focus is constant. The maximum diameter of an ellipse is the **major axis**, and half of this diameter is the **semimajor axis**. The two foci lie on the major axis; the greater their separation, the more elongated the ellipse. See also *eccentricity*, *orbit*.

elliptical galaxy A galaxy that appears round or elliptical in shape and normally contains very little gas or dust. See also *galaxy*.

elongation The angle between the Sun and a planet, or other solar system body, when viewed from Earth. The elongation of a planet is 0° when it is in conjunction with the Sun and 180° when it is at opposition. **Greatest**

elongation is the maximum possible elongation of a body, such as Mercury or Venus, that lies inside the orbit of Earth. See also *conjunction*, *opposition*.

emission line see *spectral line*.

emission nebula see *nebula*.

equatorial mounting A mounting that allows a telescope to be turned around two axes, one of which (the **polar axis**) is parallel to, and the other (the **declination axis**) perpendicular to, Earth’s axis of rotation. The telescope can follow the motion of a celestial object across the sky by being driven around the polar axis in the opposite direction of Earth’s rotation at a rate of one revolution per sidereal day. See also *declination*, *right ascension*, *sidereal time*.

equinox An occasion when the Sun is vertically overhead at a planet’s equator, and day and night have equal duration for the whole planet. In the case of Earth, the northern **vernal equinox** is the point at which the Sun crosses the celestial equator from south to north, on or around March 20 each year, and the northern **autumnal equinox** is the point at which the Sun crosses the celestial equator from north to south, on or around September 22. See also *right ascension*.

eruptive prominence see *prominence*.

eruptive variable see *variable star*.

escape velocity The minimum speed at which a projectile must be launched in order to recede forever from a massive body and not fall back. The escape velocity at Earth’s surface is 25,200 mph (40,320 km/h).

event horizon see *black hole*.

extrasolar planet (exoplanet) A planet that revolves around a star other than the Sun.

F

facula (plural: faculae) A patch of enhanced brightness on the solar photosphere that may be seen in a white-light image of the Sun, usually near the edge of the Sun’s visible disk where the background brightness is lower. Faculae correspond to regions that are hotter than their immediate surroundings. They are associated with active solar regions but may appear before, and persist after, any sunspots that develop in those regions. See also *photosphere*, *sunspot*.

Fraunhofer line One of the 574 dark absorption lines in the spectrum of the Sun that were identified by the 19th-century German optician and instrument-maker Joseph von Fraunhofer. See also *spectral line*.

flare star A faint, cool, red dwarf star that displays sudden, short-lived increases in luminosity caused by extremely powerful flares that occur above its surface. See also *red dwarf star*.

flat universe A universe in which the overall net curvature of space is zero.

In such a universe, space is flat in the sense that, apart from localized distortions caused by massive bodies, its large-scale geometry is Euclidean and light rays travel in straight lines. A universe will be flat if its overall average density is equal to a particular value, called the **critical density**. See also *closed universe*, *open universe*, *oscillating universe*.

focal length The distance between the center of a lens, or the front surface of a concave mirror, and the point at which it forms a sharp image of a very distant object.

frequency The number of wave crests of a wave motion that pass a given point in one second. In the case of an electromagnetic wave (for example, light) the frequency is equal to the speed of light divided by the wavelength. See also *electromagnetic radiation*.

fusion (nuclear fusion) The process by which atomic nuclei are joined together during energetic collisions to form heavier atomic nuclei, with an associated release of large amounts of energy. Stars are powered by fusion reactions that take place in their central cores. In a main-sequence star such as the Sun, fusion reactions convert hydrogen into helium. See also *main sequence*.

G

galactic cluster see *open cluster*.

galaxy A large aggregation of stars and clouds of gas and dust. Galaxies, which may be elliptical, spiral, or irregular in shape, contain from a few million to several trillion stars and have diameters ranging from a few thousand to over a hundred thousand light-years. The Sun is a member of the **Milky Way galaxy**, which is also sometimes known as **the Galaxy**. See also *Milky Way*.

galaxy cluster An aggregation of galaxies held together by gravity. Clusters that contain up to a few tens of member galaxies are called groups. Larger clusters are divided into **regular** and **irregular clusters**, depending on their degree of structure. The most richly populated regular clusters (**rich clusters**) contain up to several thousand galaxies.

galaxy supercluster A cluster of galaxy clusters, which is a loose aggregation of up to about ten thousand galaxies, spread through a volume of space with a diameter of up to about 200 million light-years. See also *galaxy cluster*.

Galilean moon One of the four largest natural satellites of the planet Jupiter, which were discovered in 1610 by the Italian astronomer Galileo Galilei. In order of distance from the planet, they are Io, Europa, Ganymede, and Callisto.